

How to give read and write permissions using Unix commands.
A very simple example is to create a text file with a C program,
then use chmod.

Create a file called permission.c

```
#include <stdio.h>

int main()
{
    FILE *fp;
    fp = fopen("sample.txt", "w");
    if (fp == NULL)
    {
        printf("Cannot open file\n");
        return 1;
    }
    fprintf(fp, "Hello, this is a test file.\n");
    fclose(fp);
    printf("File created successfully.\n");
    return 0;
}
```

Compile

```
gcc permission.c -o permission
```

Run

```
./permission
```

Output

File created successfully.

A file called sample.txt is created.

Check the current permissions

```
ls -l sample.txt
```

You may see

```
-rw-r--r-- 1 user user 31 Aug 17 08:20 sample.txt
```

The owner has: Read + Write

Group has: Read

Others have: Read

This is permission 644.

Give Read and Write permission to everyone

```
chmod 666 sample.txt
```

Check `ls -l sample.txt`

You should see

```
-rw-rw-rw- 1 user user ... sample.txt
```

Instead of numbers, you can directly give:

```
chmod u+rw sample.txt
```

```
chmod o+rw sample.txt
```

```
chmod a+rw sample.txt
```

Task of the Day

- 1) Give Read and Write only to the owner
- 2) Give Read and Write to the owner and group
- 3) Only owner can read and write
- 4) Owner read/write, everyone else read-only
- 5) Owner can do everything, others can only read and execute
- 6) Owner and group can read/write
- 7) Only owner can execute

Fill in the permission numbers for the following:

- 8) Owner: read + write; Group: nothing; Others: nothing
→ 600
- 9) Owner: read + write; Group: read; Others: read → 644

- 10) Owner: read + write + execute; Group: read + execute;
Others: read + execute → 755
- 11) Owner: read + write; Group: read + write; Others:
nothing → 660
- 12) Everyone: read only → 000

```
prajwal@PRAJWAL:~$ ls
buffer  buffer.c  copy.txt  hello  hello.c  os  sample.txt
prajwal@PRAJWAL:~$ cd os
prajwal@PRAJWAL:~/os$ ls
fork  q1  q1.c  q2  q2.c  q3  q3.c
prajwal@PRAJWAL:~/os$ nano permission.c
prajwal@PRAJWAL:~/os$ gcc permission.c -o permission
prajwal@PRAJWAL:~/os$ ./permission
File created successfully.
prajwal@PRAJWAL:~/os$ ls -l sample.txt
-rw-r--r-- 1 prajwal prajwal 28 Aug 17 05:22 sample.txt
prajwal@PRAJWAL:~/os$ chmod 600 sample.txt
prajwal@PRAJWAL:~/os$ ls -l sample.txt
-rw----- 1 prajwal prajwal 28 Aug 17 05:22 sample.txt
prajwal@PRAJWAL:~/os$ chmod 644 sample.txt
prajwal@PRAJWAL:~/os$ ls -l sample.txt
-rw-r--r-- 1 prajwal prajwal 28 Aug 17 05:22 sample.txt
prajwal@PRAJWAL:~/os$ chmod 755 sample.txt
prajwal@PRAJWAL:~/os$ ls -l sample.txt
-rwxr-xr-x 1 prajwal prajwal 28 Aug 17 05:22 sample.txt
prajwal@PRAJWAL:~/os$ chmod 660 sample.txt
prajwal@PRAJWAL:~/os$ ls -l sample.txt
-rw-rw---- 1 prajwal prajwal 28 Aug 17 05:22 sample.txt
prajwal@PRAJWAL:~/os$ chmod 300 sample.txt
prajwal@PRAJWAL:~/os$ chmod 000 sample.txt
prajwal@PRAJWAL:~/os$ ls -l sample.txt
----- 1 prajwal prajwal 28 Aug 17 05:22 sample.txt
prajwal@PRAJWAL:~/os$ |
```